

Peter Huang

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Education

McGill University: *Bachelor's of Engineering in Computer Engineering* Sept 2024 - May 2028
• Relevant Coursework: Data Structures & Algorithms, Software Systems, Operating Systems, Computer Organizations, Software Engineering, Object-Oriented Programming
• 2026 McHacks First Place Overall
• GPA: 4.0 / 4.0

Marianopolis College (CEGEP): *Diploma of Collegial Studies in Honours Pure & Applied science* Sept 2022 - May 2024
• **Member:** Marianopolis SFAM Club, Marianopolis Flag Football

Certifications:

AWS Solutions Architect Associate June 2025 - June 2028
AWS Solutions Architect Professional July 2025 - July 2028

Work Experience

Hedge Fund Programmer Intern | Drakkar Capital Jan 2026 - Aug 2026

Tech: Python, C, Lua, Wireshark, XMT, Scapy, PCAP,

- Implemented a **Lua-based PCAP dissector in Wireshark** to extract and decode stock trade information from captured PCAP network packets, streamlining analysis of financial data streams.
- Data pipeline design & implementing general functions and medallion architecture.

Data Engineer Intern | Transport Canada Sept 2025 - Dec 2025

Tech: Azure Databricks, Azure Data Factory, Azure DevOps, Power BI, Git, SQL, Python

- Transformed million-scale raw data in varied formats (docx, pdf, html) using **PySpark**, delivered production **Power BI** dashboards for business intelligence and data analytical insights.
- **Maintained infrastructure** including Analytics Workspaces, Lakehouse, and Purview, engineered **ETL pipelines** using Azure Databricks and Azure Data Factory, and managed Git-based CI/CD workflows in Azure DevOps.
- Designed a **GenAI-powered data ingestion** pipeline that transformed messy, multi-source data (PDFs, databases, unstructured text) using LLM-driven categorization, translation, deduplication, and RAG-based evaluation, with a focus on cost efficiency.
- Innovated an **ETL pipeline** for **docx** ingestion, redesigning **string-based parsing** into a **structured JSON-like** process, improving efficiency and reducing processing time by over **90%**.

AI Developer Intern | CloudAct CPA Aug 2025 - Sept 2025

Tech: NLP, LLM, Word Embeddings, PostgreSQL, Render, RAG, LangChain, BeautifulSoup

- Developed automated LLM pipeline to monitor **SEC RSS feeds** and extract information from filings (Extraction S-1 filings reached 90% accuracy).
- Implemented **document chunking** and **semantic chunk matching** using word embeddings, designed formatted output retrieval.
- Integrated **PostgreSQL** for structured data storage. Deployed scalable and reliable availability system on **Render**.

Cloud Architect Intern | Coinchain Inc. May 2025 - Aug 2025

- Architected end-to-end **AWS Cloud solutions**, optimizing cost, performance, security, and scalability using VPC design (public/private subnets, NAT gateways, Internet gateways), EC2, S3, RDS, IAM, CloudFront, and Route 53.
- Supported companies in listing and managing SaaS, AMI, and container products on the **AWS Marketplace**, including setup of billing, metering, and deployment configurations.

Collegial Level Tutoring | Marianopolis College & Private Tutor Sept 2023 - Current

- Conducted one-to-one tutoring sessions for 10+ students in Calculus, Mechanics, E&M, and Computer Programming.

Research Experience

Student Researcher | McGill AI Lab Aug 2025 - Current

Tech: NLP, Word Embeddings, Web Scraping, Data Processing

- Applied large data ETL from **HuggingFace** and word embeddings (**Word2Vec**, **roBERTa**) to implement sentiment analyze news articles on Israel-Palestine conflict bias.

Projects

McHacks 1st Place Winning Project — McGill Go | [devpost](#)

- Full-stack project that improves McGill student experience for exchange program researching, deployment to AWS soon.

ITrackApply Google Extension | [youtube demo](#) | Python, GoogleSheets API, AWS, OAuth 2.0, OpenAI

- Developed a **Chrome extension** that automates job application tracking with **AWS** integration.
- Designed simple and easy-to-use interface using **GoogleSheets API**.
- Implemented secure **OAuth 2.0** authentication and real-time synchronization to **DynamoDB**.
- Built a scalable Python Flask backend deployed on **AWS Lambda** and integrated with **API Gateway**.
- Leveraged LLM-powered data extraction, focused on cost optimization techniques.

ScrapeNYT | [github](#) | *Python, API, Newspaper4k*

- Scrapes 2500+ payload articles per hour from New York Times using **Newspaper4k** and records in JSONL file for further ML purposes.

Cybersecurity & Systems Programming | [github](#) | *C, Bash, Cybersecurity*

- Developed dynamic memory, async, low-level network servers and implemented prompt injection, buffer overflow offensive cybersecurity techniques in **C**, automated system tasks using **Bash**.

AI MoodDetector | [github](#) | *ML, PyTorch, OpenCV*

- Trained and fine-tuned ResNet-152 Model on emotion image dataset with **PyTorch**, implemented live camera integration with **OpenCV**.

Grocery Store GUI Application | [github](#) | *Java, Umple, JUnit, VSCode debugging tools*

- Developed a full-stack grocery store management application as a group project using **Java** with a **Model-View-Controller** architecture.
- Designed system models with Class Diagrams and State Machines using **Umple**.
- Built responsive UI components using **SceneBuilder** and conducted tests using **VSCode debugging tools**.

Warehouse Management System | [github](#) | *React, Node.js, MongoDB*

- Built a **full-stack warehouse management system** using **MERN** stack, enabling real-time inventory tracking, user authentication, and data persistence.

Skills

Programming Languages: Python, Java, C, Lua, SQL, VHDL, Bash Shell

Databases: SQL, PostgreSQL, noSQL (MongoDB), DynamoDB

Tools: Microsoft Azure, AWS, Git, Unity, Power BI, WireShark

Languages: English, French, Chinese (Mandarin)

Achievements

- 2026 McHacks First Place Overall
- 2025 McGill John Howard Ambrose Scholarship
- 2025 McGill Dean's Honour List
- 2024 Marianopolis Dean's List
- 2023 RSEQ Flag Football All-Star Award
- 2022 Football Most Polyvalent Player Award
- 2022 Physics Academic Excellence Award